

Valeri Perepetchai
190 112TH Ave N
Saint Petersburg, FL 33716
Res: (772) 404-4832 ✉ valeri.perepetchai@mail.mcgill.ca

Summary

Senior Quantitative Analyst/Software Developer with 10 years of experience in various industries. Developed an expertise in artificial intelligence, machine learning, natural language processing, statistical analysis, digital signal processing, genetics, cryptography, computational fluid dynamics and many other fields by using C/C++/C#/Java/Matlab/Python. Recognized as being creative, innovative, and autonomous, with a strong sense of innovation, while possessing exceptional analytical and problem solving skills.

Main competencies:

- Deadline management
- Project coordination
- Client needs analysis
- Research and development
- Training and management
- Report preparation and presentations

Main accomplishments:

- COVID-19 detection in saliva spectra with 99% of accuracy by machine learning.
- Created artificial intelligence system for insider threat detection. Tested on CERT data (<ftp://ftp.sei.cmu.edu/pub/cert-data>) with no missed detection and a false alarm less than 2%
- Completed 3 patents devoted to data mining and pattern recognition of big data sets.
<https://patents.google.com/?inventor=perepetchai&oq=perepetchai>
- Developed GPS positioning with cm-level accuracy and the statistical algorithms, detecting outliers in satellite data in real time and meeting the navigation requirements of U.S. Department of Avionics for Canadian Marconi Co. http://digitool.library.mcgill.ca/R/-?func=dbin-jump-full¤t_base=GEN01&object_id=29400

Computer Skills:

- C/C++/C#, Java, Matlab, Stata, R, SAS, Fortran, Python, Perl, PowerShell
- SQL, relational database structures, Hadoop tools, big dataset use
- UNIX (shell programming, gcc, g++, make), Linux, WindowsNT, DOS
- Multithreading programming

Professional Experience

KALIGIA BIOSCIENCES, Largo, FL

2019/09- 2020/10

Senior Quantitative Analyst

- Developed machine learning algorithms to detect COVID-19 in saliva spectra. Filled for FDA approval.
- Created artificial intelligence and machine learning algorithms (regression, classification, clustering, deep learning) for little invasive blood analysis in Python/Java/TensorFlow

VERIATO (SPECTORSOFT), Vero Beach, FL

2014/12- 2019/07

Senior Software Engineer

- Created artificial intelligence and machine learning algorithms in C# (kernel non-parametric probability density estimation, SVM, nonlinear PCA, logistic regression, NIPALS, neural networks, spectral clustering, OPTICS and etc.) for user-entity behavior analysis: insider detection, continuous user identification, user bio-metric authentication (password hardening), malware and network attack detection.
- Developed NLP algorithms in C# for sentiment analysis and topic identification (LDA) for security monitoring in social nets (Twitter).
- Collected and forwarded event logs in PowerShell for UBA.
- Developed Optical Character Recognition in C#.

- Developed in C# a sniffer and intrusion detection system for network security monitoring and full packet analysis.

SELF-EMPLOYED

2009/01- 2014/12

- Developed quantitative price models using data mining and natural language processing of social nets in Java. Developed sentiment based trading strategies and automatic equity trading systems in C++ under Linux with bash shell programming. Multithreading programming in real time. Tested the developed trading strategies on real time stock data in www.investopedia.com.

ISentium Technology Hedge Fund, Montreal

- Dental 3D computer vision software development (edge detectors, surface segmentation, mesh triangulation) in C++/OpenGL under Windows. ***Evirsa Inc./Cynoprod Inc., Montreal.***
- Quantitative trait analysis. Statistical analysis of large dataset (Stata, Python, Excel/VBA, Hadoop tools, UNIX clusters with shell programming). ***Jewish General Hospital, Montreal.***

GENIZON BIOSCIENCES, Montreal

2003/02- 2009/01

Senior Researcher

- Developed statistical algorithms for genome – wide association analysis in genetics with C/C++ under WindowsNT, Unix clusters with shell programming (make, gcc, g++) by using Markov chain Monte Carlo simulations, data mining, pattern recognition, neural networks, numerical optimization techniques, decision trees, CART and random forest.
- Created a patent-pending algorithm for the correction of population stratification and diagnostics with C++ (STL) under Linux/WindowsNT by using unsupervised machine learning techniques (PCA, hierarchical clustering and intelligent K-means algorithm).
- Created a patented statistical algorithm for the analysis of pooled DNA samples with C++/MATLAB/SAS/R under Linux/WindowsNT .
- Multithreading programming in grids, big dataset use.
- Completed 6 scientific papers and 3 patents.

ORTHOSOFT LTD., Montreal

2001/04-2002/11

Senior Software Developer

- Created algorithms improving accuracy of optical real-time 3D tracking systems in 10 times (DSP, time series, stochastic model estimation, signal de-noising, statistical analysis).
- Designed a patent-pending IR optical tracker reducing the cost of computer-aided surgery.
- Elaborated 3D computer vision and image processing for medical applications (MATLAB, C++, Open GL, UNIX).

Education

M.S. in Computer Science

2000

McGill University, Montreal

M.S. with honors in Solid State Physics and Electronics

1996

Institute of Electronic Engineering, Moscow

Status: Canadian citizen under H1B visa (10/01/2017 – 09/08/2022)